

Problem Set #7: Classical Mechanics (PH31207)

October 8, 2026

Hamilton's Canonical Equations

1. Consider the following Hamiltonian corresponding to the three site Fermi-Pasta-Ulam-Tsingou chain with periodic boundary conditions :

$$H(p_1, p_2, p_3, x_1, x_2, x_3) = \frac{1}{2}(p_1^2 + p_2^2 + p_3^2) + V(x_2 - x_1) + V(x_3 - x_2) + V(x_1 - x_3) \quad (1)$$

Where :

$$V(x_{i+1} - x_i) = \frac{1}{2}(x_{i+1} - x_i)^2 + \frac{1}{12}(x_{i+1} - x_i)^3 \quad (2)$$

- (a) Derive all the canonical equations of motion.
 - (b) Using these equations show that $p_1 + p_2 + p_3$ is conserved.
2. The Lagrangian for a certain one-dimensional motion is

$$\mathcal{L}(x, \dot{x}, t) = \frac{e^{\gamma t}}{2} (m\dot{x}^2 - kx^2),$$

where γ , m , and k are positive real constants.

- (a) Find the canonical momentum $p = \partial\mathcal{L}/\partial\dot{x}$.
 - (b) Construct the Hamiltonian $\mathcal{H}(x, p, t)$.
 - (c) Is the Hamiltonian a constant of motion?
3. Two inert gas atoms, each of mass m , interact classically via the potential

$$V(r) = -\frac{2A}{r^6} + \frac{B}{r^{12}}, \quad A, B > 0, \quad r = |\vec{r}_1 - \vec{r}_2|.$$

Describe this problem using the center-of-mass coordinate $\vec{R} = (\vec{r}_1 + \vec{r}_2)/2$ and the relative coordinate $\vec{r} = \vec{r}_1 - \vec{r}_2$, with reduced mass μ .

- (a) Obtain the Hamiltonian of this system.
- (b) Describe the lowest-energy classical state(s) of this system.